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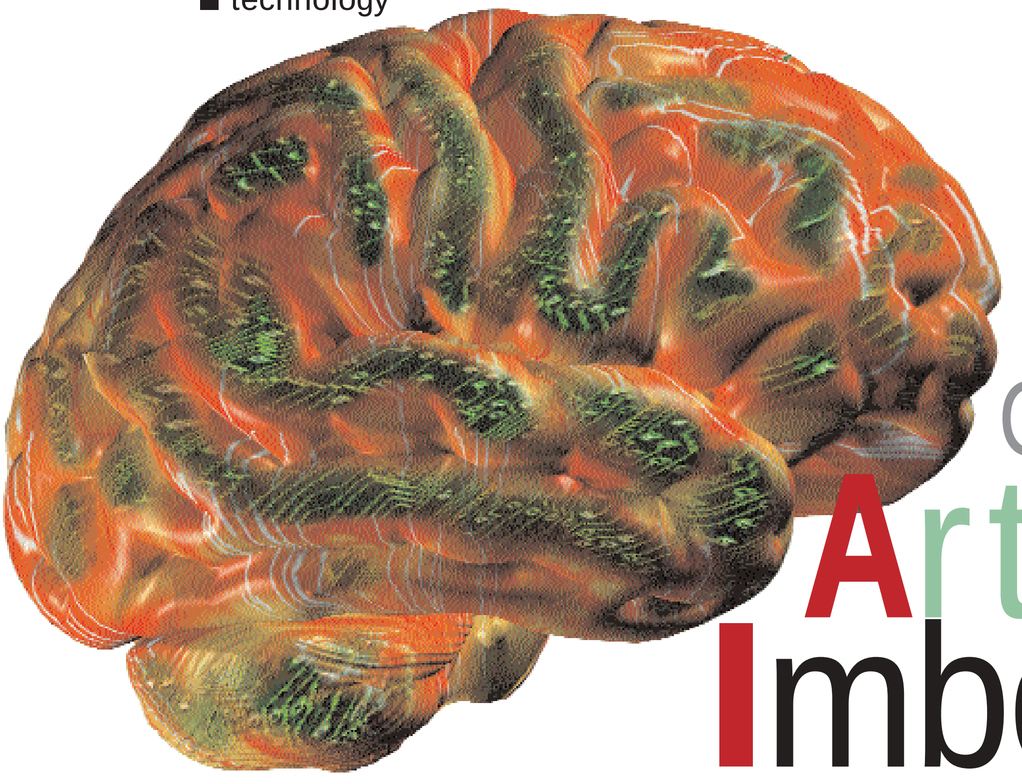

CREATING Artificial Imbeciles

"It's the year 2000, but where are the flying cars? I was promised flying cars."
— Avery Brooks, IBM commercial

The term AI spans everything from 'The Logic Theorist'—a simple theorem-proving program, to the scary idea that we might be ruled by an army of robots. It's interesting to know what comes to most people's minds when they hear the word AI: movies, smart robots and games. Generally, the idea is of intelligent machines, or machines that think, talk and so on. In fact, that idea is pretty close to what the pioneers of the field known as Artificial Intelligence imagined. But AI means different things to different people. To some, AI is not a given thing; they ask whether AI will be possible, that is, whether a machine can ever be truly intelligent. To others, the answer is a definite Yes, and their questions are about how to infuse a machine with intelligence. There are others who couldn't care less, and simply build systems that do intelligent things.

So what is AI? There are three important aspects to that question. The first is about whether machines can think, and why is it difficult to create an intelligent machine. The second is about AI as a technology. And the third is the rather interesting history of the field—its ups and downs, shifts of focus and its sometimes over-optimistic predictions.

Perspective

As far as history goes, AI is not, and never was, a single technology at all. Coined in 1956 by John McCarthy, the term indicated intelligent behaviour by machines, but the term has encompassed many technologies. At the backdrop of it all is Marvin Minsky—the undisputed champion of AI; a scientist who has always maintained that a brain is nothing but a machine, and who's always saying that we'll have super-robots soon (*see box 'Minsky's Mind-Brain Musings'*). Whether that happens or not is, naturally, open to debate.

In 1943, Warren McCulloch and Walter Pitts, both computer scientists as well as mathemeticians, found that a network of artificial neurons could model some functions of the brain. That event started off the field of Artificial Neural Networks (ANN). ANNs are different from regular intelligent systems, because they can learn. The Dartmouth Conference that was held in 1956, can be called the event that gave birth to AI. Proposals for how a computer might be taught to use a language, how a machine might learn, how it might get to be creative and so on, were laid out.

Between 1956 and 1966, several primitive AI programs such as 'Student'—a program that could solve algebra-type word problems, and 'The Logic Theorist', which could prove mathematical theorems, were developed. The enthusiasm for the new field at that time was such that renowned computer scientist, Herbert Simon stated that by 1985, machines would be capable of doing anything a person could do.

In 1967, however, Minsky and Seymour Papert—who, with Minsy, was co-founder of the AI Lab at MIT—showed that ANNs could not solve certain pattern

The Story So Far...

Honda's Asimo humanoid robot moves freely around, and can climb and descend stairways and slopes. It can listen to and guide customers.

In 1995, an automatic vision system steered a vehicle across the United States.

Face recognition technology has made it possible to monitor large crowds from a central location, as was done in Florida.

AI has made human inventions duplicable—state-of-the-art patents have been re-engineered by AI agents.

In 1999, an AI agent ran a satellite beyond Mars for over a day without ground control.